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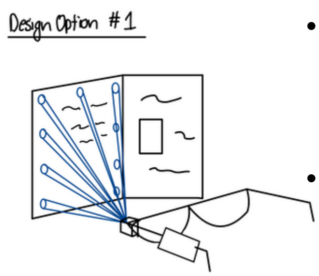
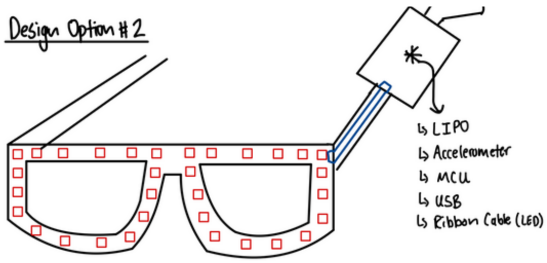
MOTION CUES GLASSES

Problem Identification

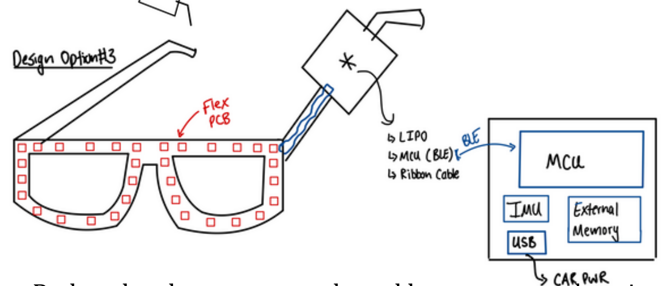
- Noticed car sickness while reading magazine on a road trip and looked into designs better than existing tacky versions
- Looked to create a solution while being heavily inspired by the Vehicle Motion Cues by Apple



- Projecting moving dots onto page required expensive hardware and complex systems like page size recognition and depth tracking
- Possibly iterate to this more complex design in future

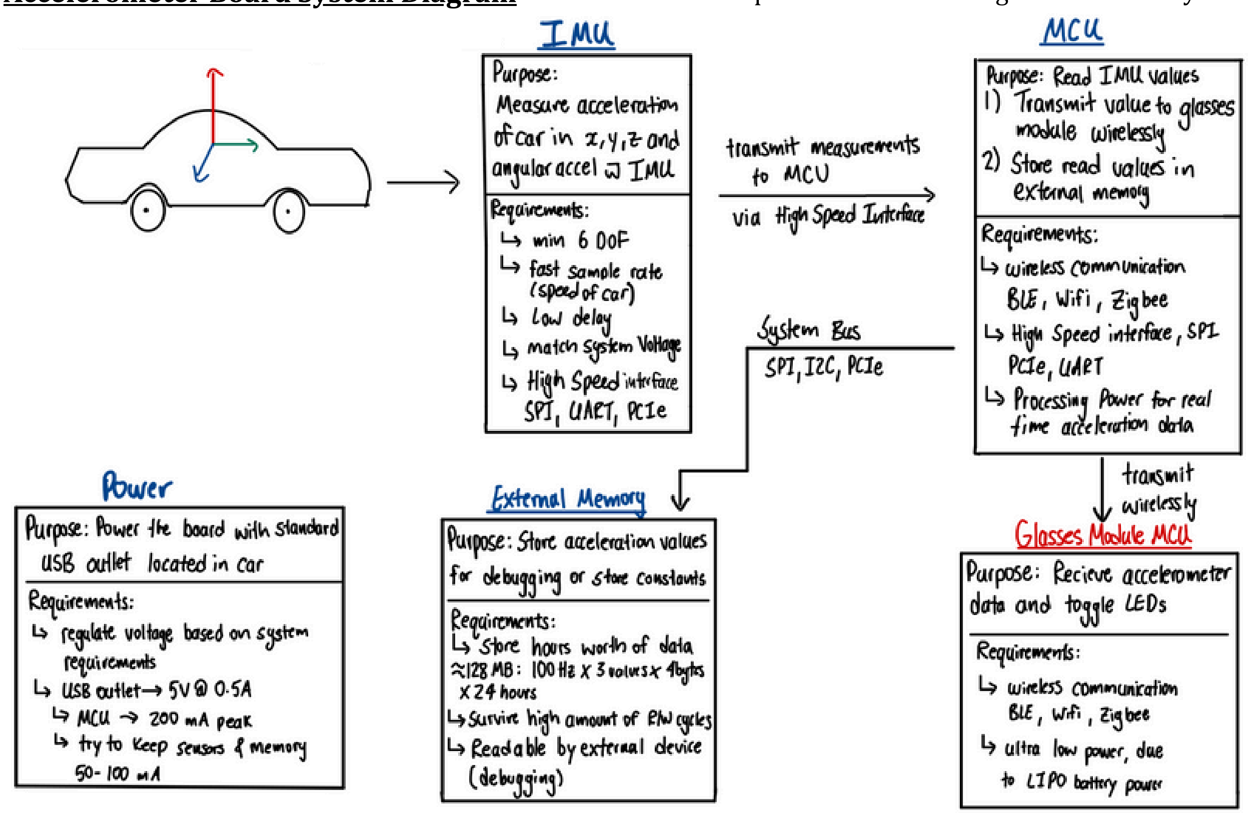


- LEDs in peripheral vision for proof of concept, more feasible
- Difficult to differentiate between car acceleration and head
- Bulky hardware on glasses temple



- Reduce hardware on temple and better car acceleration measurement with daughter PCB
- Opted for Flex PCB for glasses LED array

Accelerometer Board System Diagram



Goal: Capture Accelerometer Data of Vehicle motion and transmit to Glasses Module MCU

- Will be mounted to the floor of the moving vehicle and try to minimize any vibrations or external movement
- Selected IMU with 6DOF and decided to use SPI interface to allow for high data transfer from fast sampling
- Selected ESP32-C6 MCU for ease of integration, good wireless performance and low priority of power due to being supplied by car
- Selected Micro USB due to simple integration and not heavy use of USB features such as fast data transfer
- Decided on microSD card as it can be easily removed and read on another device and can handle many R/W cycles

Accelerometer Board Features

- 2 Layer PCB controlled by **ESP32-C6** and its USB, BLE and SPI interfaces
- Integrated Bosch BMI323 **6 axis IMU** with **16 bit** digital 3 axis accelerometer and 3 axis gyroscope
- Leverages onboard PCB antenna for possible communication with Glasses Module PCB via **BLE**
- Added external memory in the form of **microSD** via **SPI** which allows for data logging and debugging of accelerometer values
- Supports **Micro USB** for system power and for flashing ESP32-C6 module
- Uses LM3940IMPX **LDO** for power regulation of input power from 5V 0.5A USB port found in cars
- Implemented protection for physical interfaces such as USB and microSD with **protection diode** arrays
- Sized appropriate crystal capacitors for external RTC clock
- Push buttons for RESET and flashing the board
- PCB routing ensures proper **impedance matching**, minimized **signal reflections** on high speed traces, and jumper placement

Key Skills

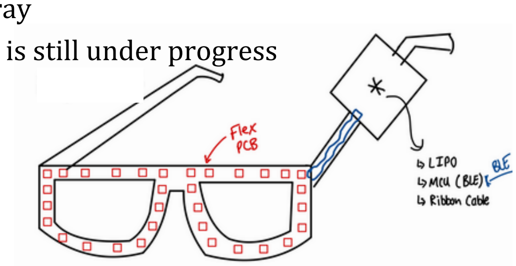
- **PCB Design (Altium):** 2 Layer board with Signal and GND layer stack up
- **Sensor Integration:** IMU sensor which communicates over SPI
- **Protocols & Interfaces:** USB, SPI, BLE, microSD

Glasses Module Board (In Progress)

Goal: To receive accelerometer data from the Accelerometer Board MCU and then light up the board accordingly

- 4 layer PCB stack up
- Small form LiPo battery due to size constraints
- Nordic nrf52840 MCU for ultra low power MCU with BLE capabilities
- Rigid Flex PCB with ribbon cable for glasses LED array

Design is still under progress



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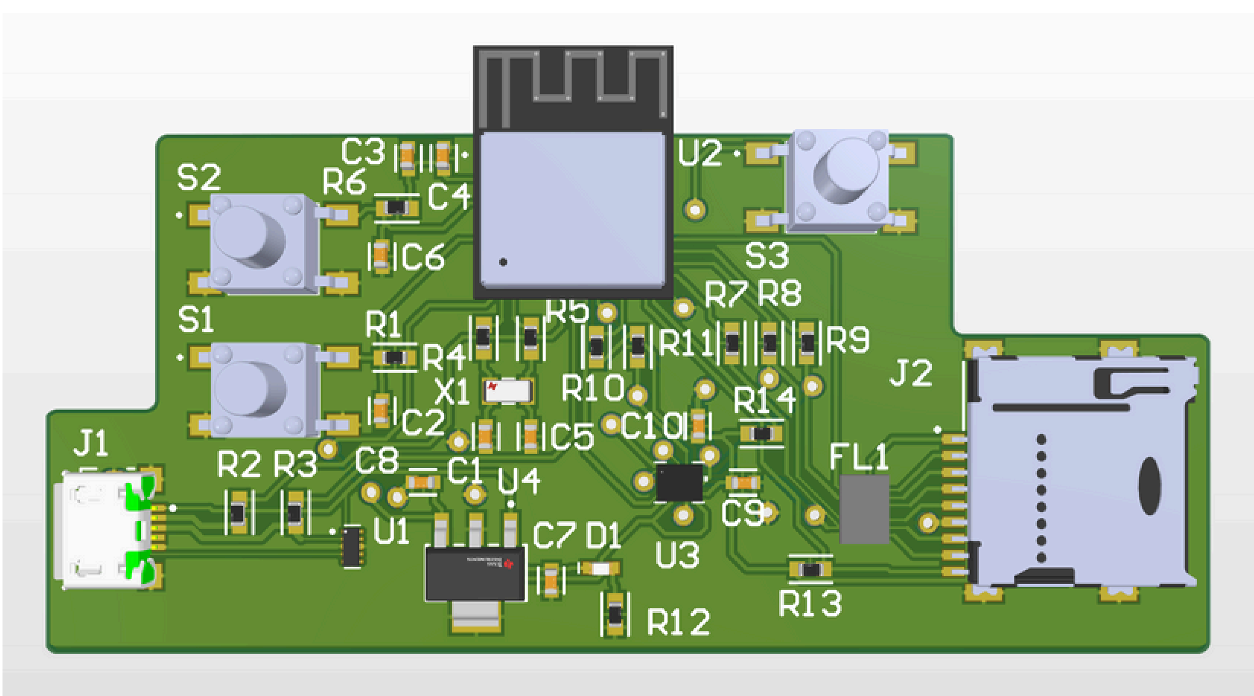
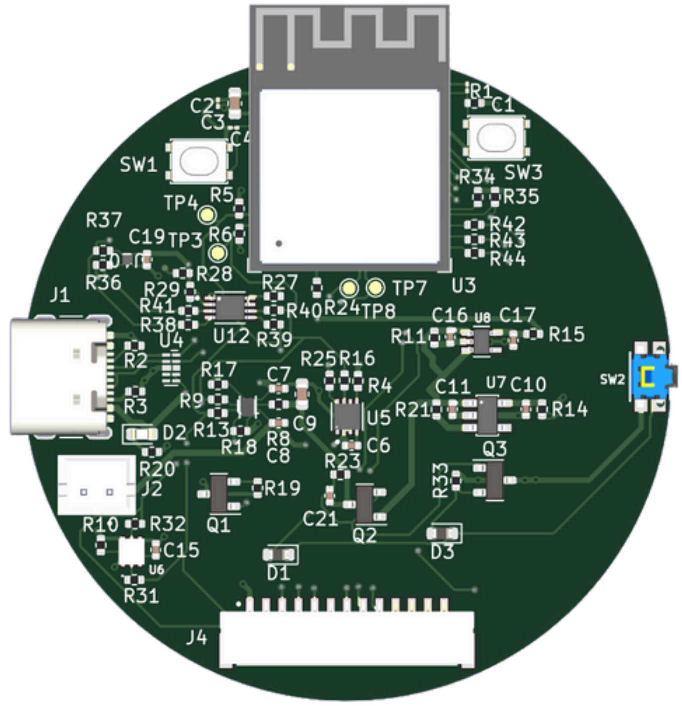


SMARTWATCH PCB PROJECT

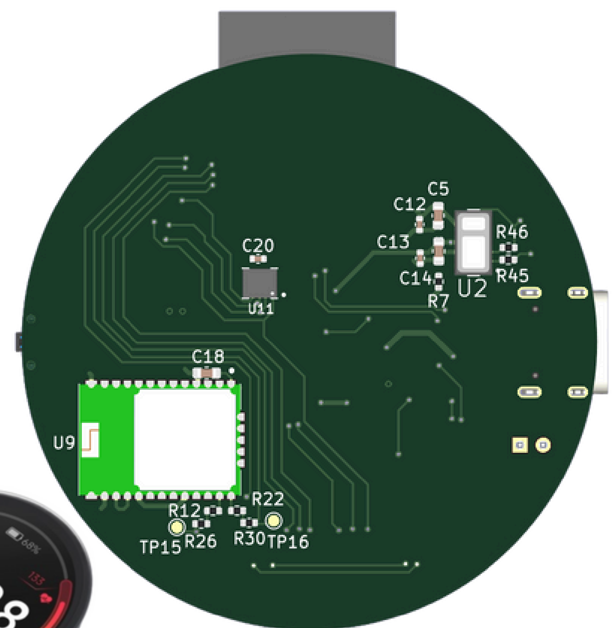
Project Goal

- Design and develop a compact, low-power smartwatch integrating multiple sensors and wireless connectivity for health monitoring and activity tracking, with efficient power management and implemented with an 1.28" circular LCD



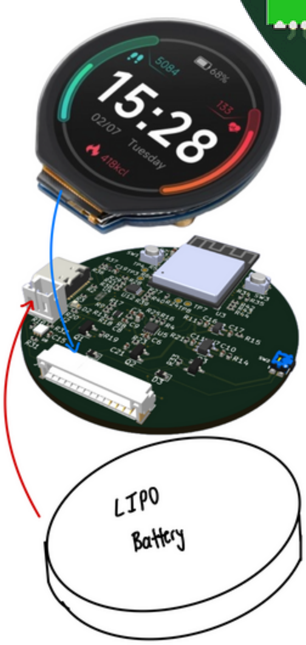
Key Skills

- **PCB Design (KiCAD)**: 4-layer board layout in KiCAD with signal integrity, power distribution, and compact routing
- **Sensor Integration**: I^2C and SPI communication with IMU, heart rate/SpO_2 sensor, GPS, compass, and ambient light sensors
- **Power Management**: USB-C charging, soft-latch power control, battery fuel gauging, and multiple voltage rails (1.8 V, 3.3 V)
- **Hardware Validation**: Bus topology design, level shifting, testpoint planning, and interface debugging
- **Protocols & Interfaces**: USB, I^2C, SPI, UART, BLE, Wi-Fi, GNSS

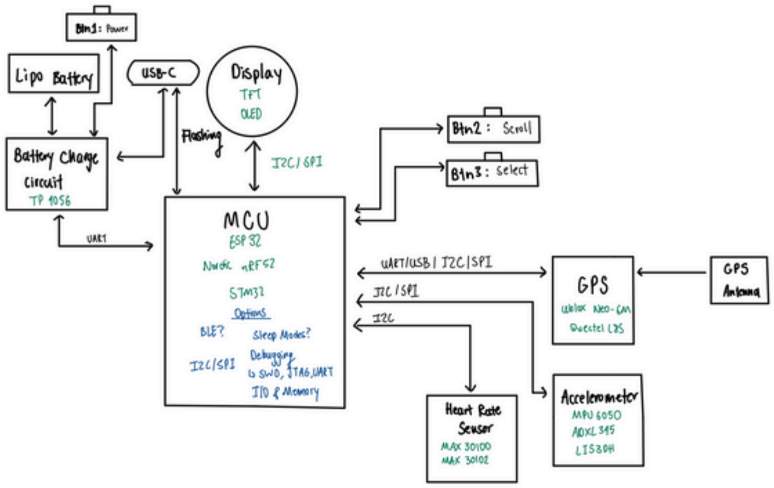


Project Features

- Designed a 1.28" circular 4-layer PCB integrating the **ESP32-S3-MINI-1** with Wi-Fi, BLE, and USB support
- Integrated sensors via I^2C/SPI for heart rate (MAX30102), IMU (LSM6DSOX), magnetometer (AK0991x), GPS (MAX-M10S / CAM-M8Q), and ambient light monitoring, with appropriate **level shifting** for 1.8 V devices
- Implemented **USB-C** power input with battery charging, **power path** management, fuel gauge, and **soft-latch** power control circuit, LDOs for regulated 3.3 V and 1.8 V power rails
- Connected a 1.28" SPI TFT LCD with capacitive touch interface for user interaction
- Focused on signal integrity, proper **test point** and **jumper** placement, and **controlled impedance** routing for high-speed and RF lines



Initial System Diagram



I2C Pull-Up Resistor Calculation/Bus Design

Max pullup calculation

	IC	Addr	V_{OL}	V_{OH}	t_{rise}
3.3V	VEML 6030	0x10	0.4	1.3	1000 ns
	MAX 17048G	0x36 (7bit)	0.5	1.4	1000 ns
	BQ 25188	0x6A (7bit) 0x D4 (8bit)	0.4	1.3	1000 ns
	GC9A01 (LCD)	0x15 (7bit)	0.6	1.4	1000 ns
1.8V	AK09918	0xC (7bit)	0.36	–	1000 ns
	MAX30102	0x57 (7bit)	0.4	–	1000 ns

$I_{OL} = 3mA$

3.3V: $R_{pullup\ min} = \dfrac{V_{CC} - V_{OL}(max)}{I_{OL}}$

$= \dfrac{3.3V - 0.6}{0.003}$

$= 900\ \Omega$

1.8V $R_{pullup\ min} = \dfrac{1.8V - 0.4}{0.003}$

$= 467\ \Omega$

traces extra

$C_B = 15pF \times 4\ devices + 20pF + 20pF$

3.3V: $R_{pullup\ max} = \dfrac{t_{rise}}{0.8473 \times C_B} = \dfrac{1\mu s}{0.8473 \times 100pF}$

$R_{pullup_{max}} = 11.8\ K$

1.8V: $R_{pullup_{max}} = \dfrac{1\mu s}{0.8473 \times 50pF}$

$= 23.8K$

3V3 LDO Junction Temperature Calculation

$V_{in} = 4.5V$

$V_{out} = 3.3V$

$I_{load} = 0.6\ A$

$I_{GND} = 8 \times 10^{-5} A$

$P_{D\,max} = (4.5V - 3.3V) \times 0.6A + 4.5V \times 8 \times 10^{-5} A$

$= 0.72\ W + 3.6 \times 10^{-4}$

$P_{D\,max} = 0.72036\ W$

Thermal Resistance (Junction to case) : $96\ °C/W$

Junction temp $= 0.72036\ W \times 96\ °C/W$

$= 69.15\ °C + 25C$

$= 94.15\ °C$ ✓

1V8 LDO Junction Temperature Calculation

$V_{in} = 4.5V$

$V_{out} = 1.8V$

$I_{load} = 0.0012\ A$

$I_{GND} = 3.9 \times 10^{-5} A$

$P_{D\,max} = (4.5V - 1.8V) \times 0.0012A + 4.5V \times 3.9 \times 10^{-5} A$

$= 3.4155 \times 10^{-3}\ W$

Junction thermal resistance: $256.5\ °C/W$

ambient

Junction temp: $256.5\ °C/W \times 3.4155 \times 10^{-3}\ W + 25C$

$= 0.876\ C + 25C$

$= 25.876C$ ✓

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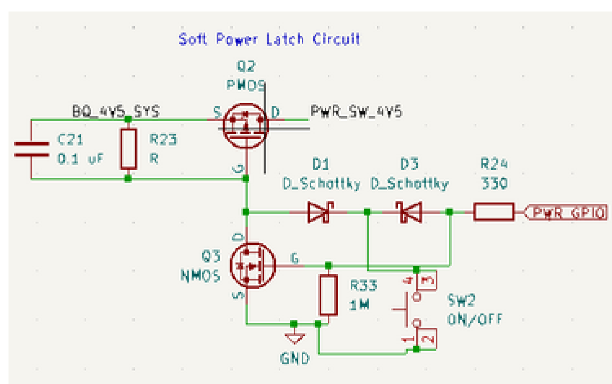
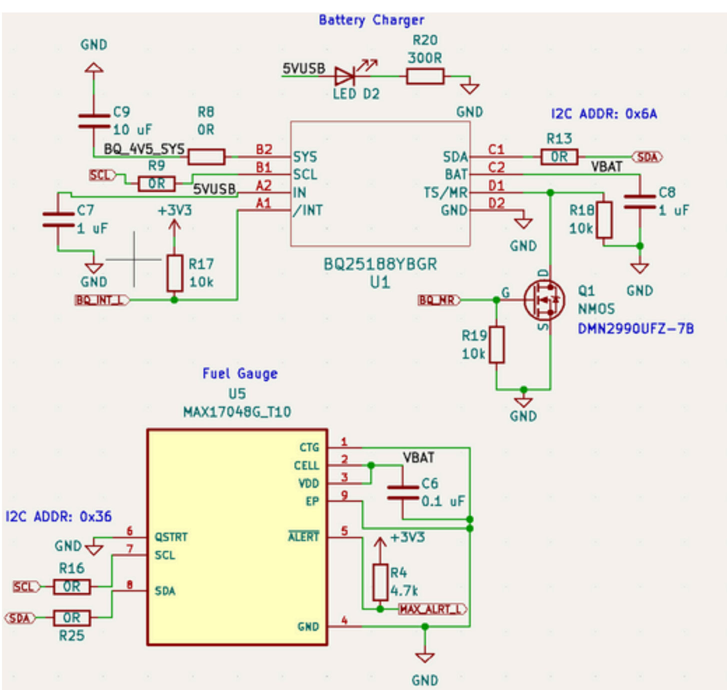
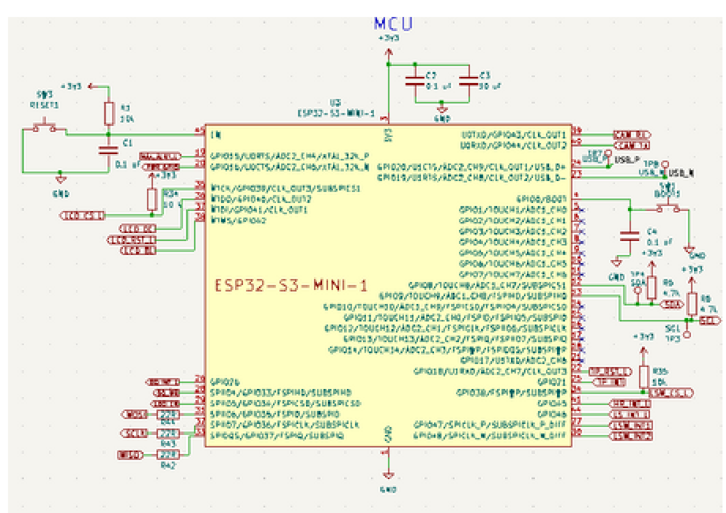
PDF

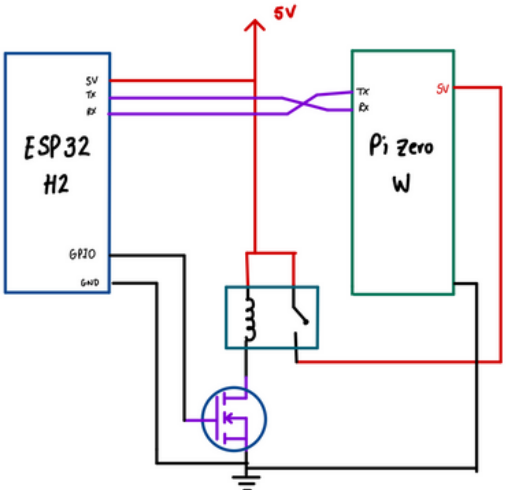


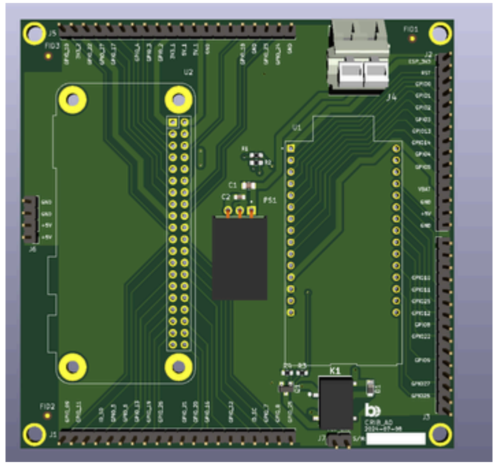


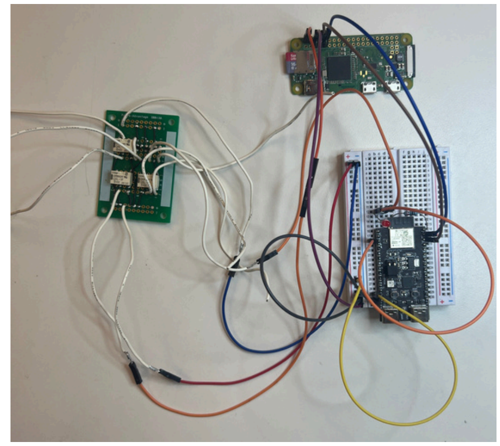


CRIB - SLEEPING PI PROJECT

brickeye







Project Goal

- Reduce the sleep current of the Pi Zero W from current 180 mA for battery powered applications
- Design for future expansion such as adding sensors, support for other MCUs
- Track time spent asleep during operation

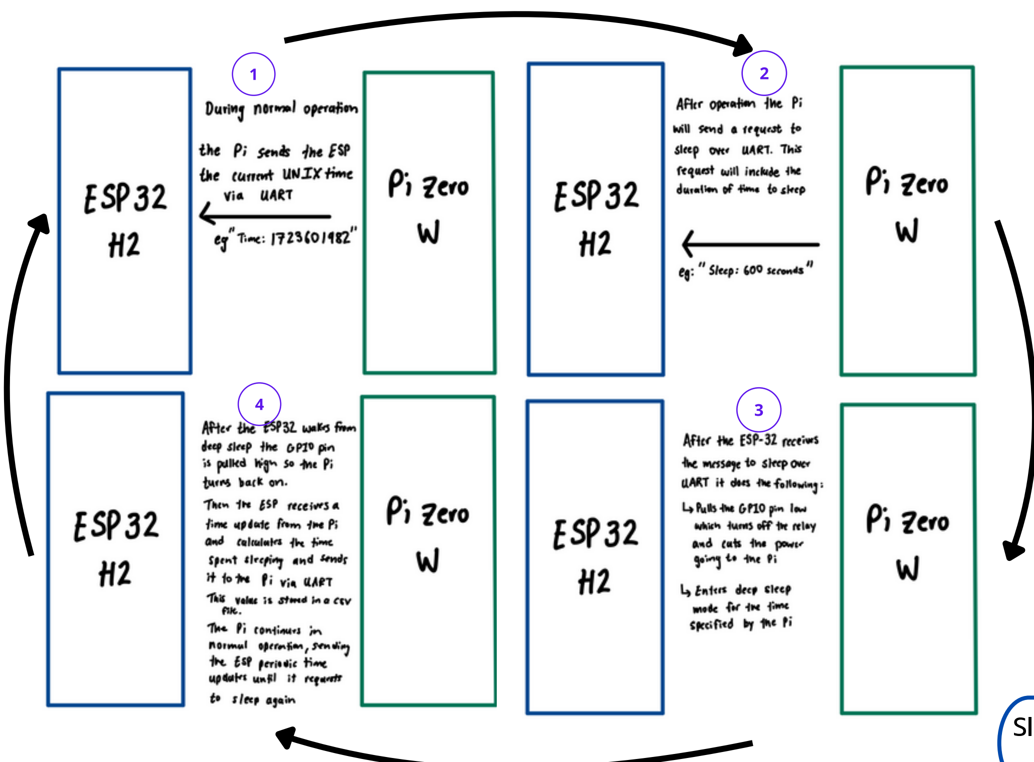
Key Skills

- PCB Design (KiCAD)
- ESP32
- UART (Software & Elec)
- C Programming
- FreeRTOS
- RTC
- Raspberry Pi
- Python

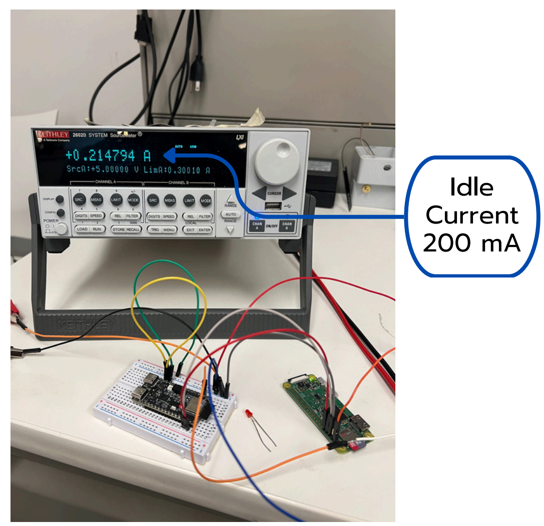
Project Features

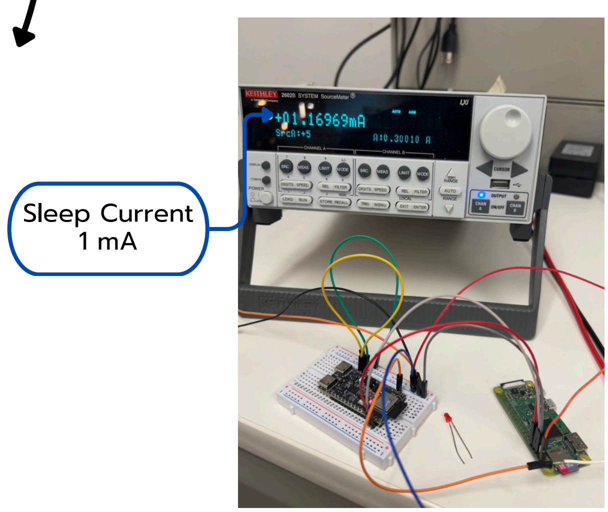
- **UART** interface between ESP and Pi Zero
- Pi power controlled by Signal Relay
- Relay controlled by **MOSFET** and ESP GPIO
- On board 24V to 5V DC Converter
- Breakout Headers for expansion
- V2 PCB features support for Adafruit M0
- Sleep Current of **1 mA**

Sequence of Operation

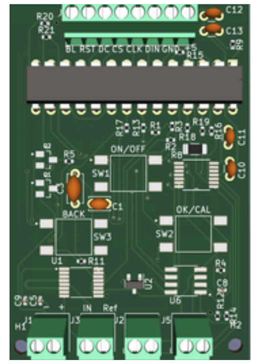


Idle vs Deep Sleep Current





MULTI-PARAMETER WATER TESTER



Project Goal

- Design a handheld device capable of measuring the parameters of a water sample such as PH, Temperature, Conductivity and ORP
- Modular design such that it can be interchanged for different applications and parameters

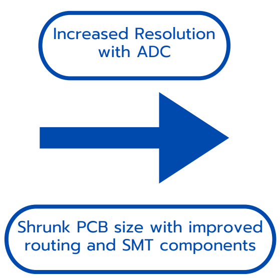
Key Skills

- Low-Voltage PCB Design (KiCAD)
- High Resolution Sensor Measurement (ADC)
- Signal Amplification (Op-Amp)
- Digital Circuits (Latch Circuit)
- SPI (LCD)
- Fusion360
- C++
- ATMega328

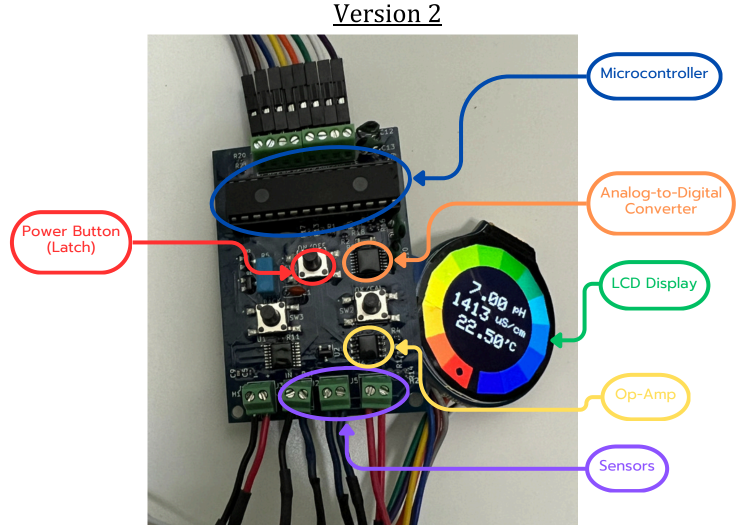
Project Features

- 2 Layer **PCB** to interface sensors with MCU and LCD
- 12 Bit 4 channel **ADC** for increased resolution
- Power latch circuit for auto-shutoff
- Op-amp circuit for PH Sensor amplification
- Custom GUI to display measurements
- 3D printed enclosure for prototype

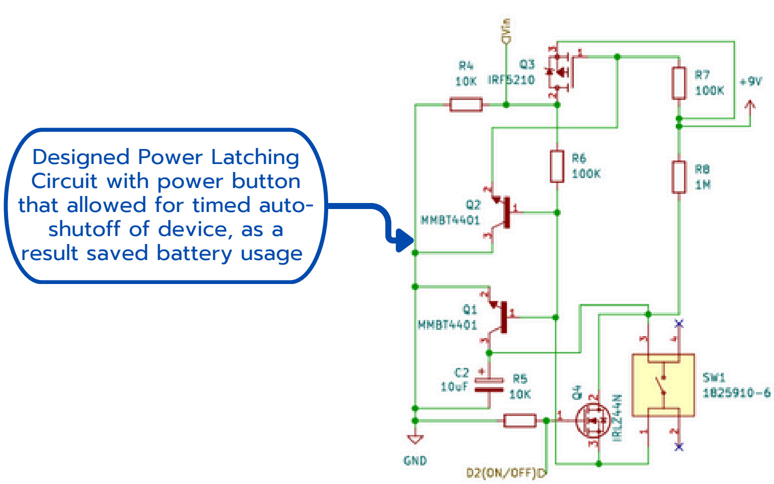
Version 1



Version 2

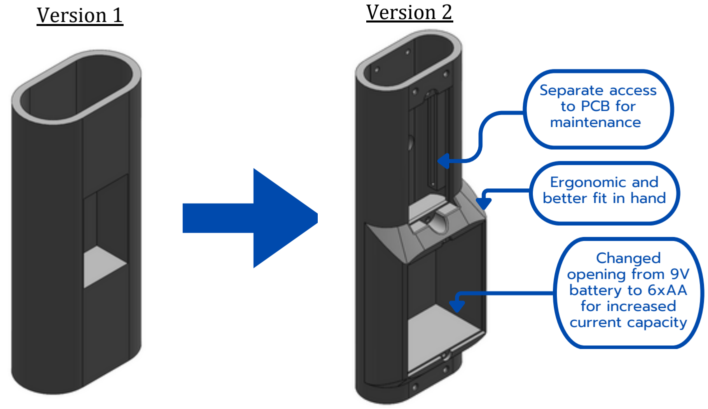


Power Latch Circuit

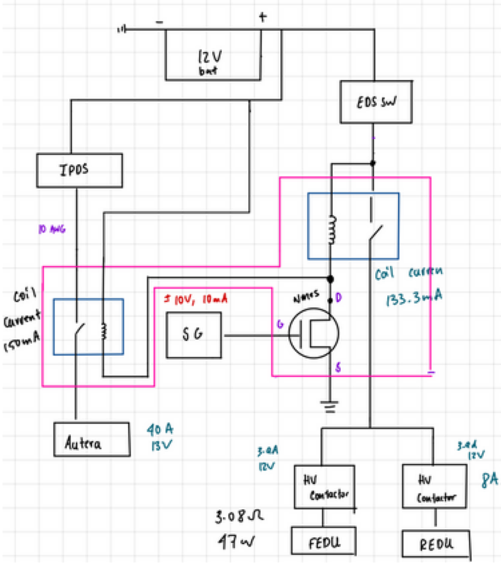


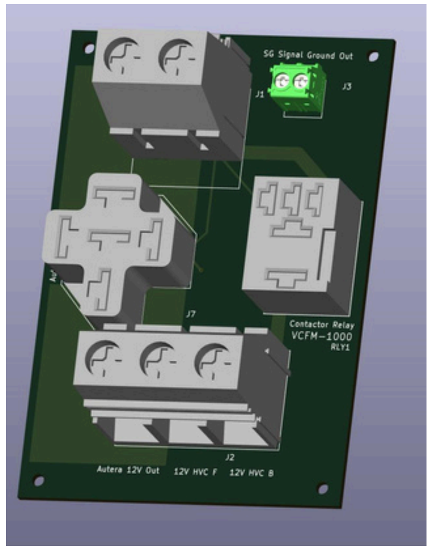
Version 1

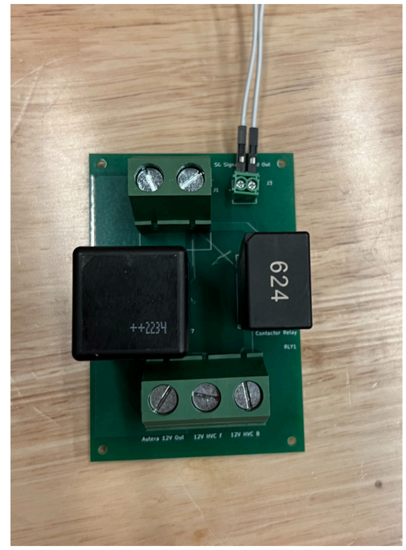
Version 2



HV CONTACTOR PCB UWAFT







Project Goal

- Design **PCB** capable of switching the power to the Autera and HV Contactors based on **3.3V signal** from Speedgoat
- Be able to handle the max current draw of **40A** and **8A** respectively

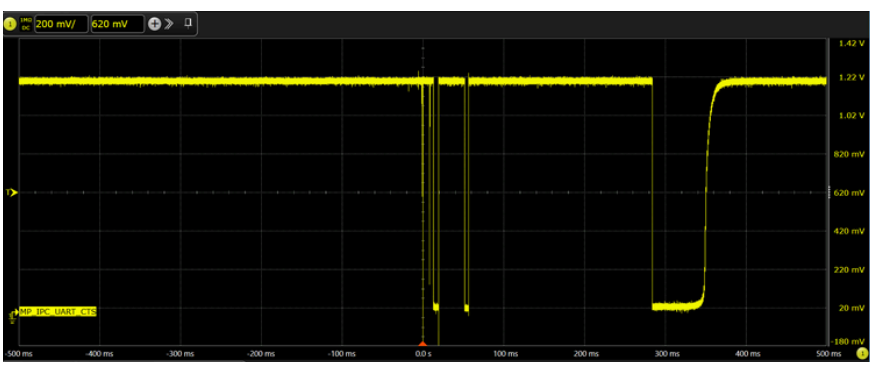
Key Skills

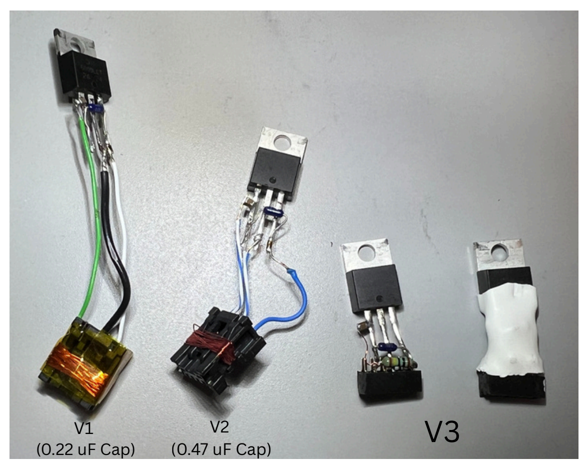
- High Current PCB Design (**KiCAD**)
- High Capacity Power Relay
- MOSFET Relay Control

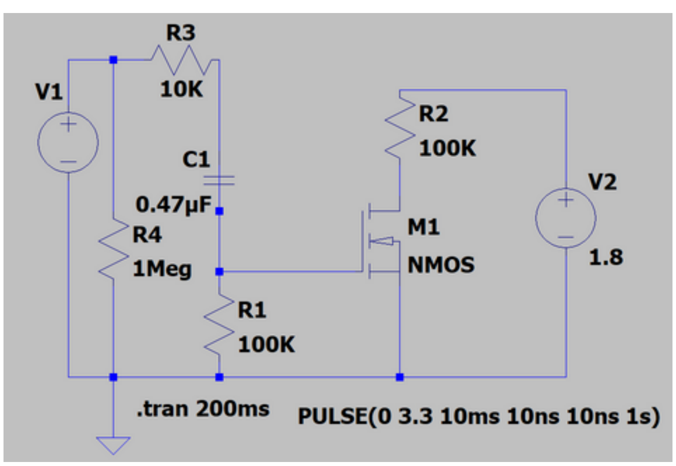
Project Features

- Automotive grade **Relays** to control Power going to Autera and Contactors
- Large terminal blocks for 10 AWG wire
- **nMOS** to control relays from Speedgoat input

SIGNAL PULL DOWN DEVICE







Project Goal

- Design and create device capable of pulling 1.2V PS_HOLD signal low for > 50ms forcing device to RAM dump
- Eliminates the need SW team from having to manually short this signal on header with wire which is super unreliable and hard to time
- Minimize size and components (use existing, iterated from 2 FET design to 1

Key Skills

- SPICE Simulation
- Prototyping
- Soldering
- Signal Capture (Scope)

Project Features

- **RC** High Pass Filter to allow transient voltage for short controlled turning on of FET
- 1M bleed resistor to fix discharging issues
- Integrated into 2mm pitch header